

Prevalence of optic pathway glioma in NF1: a systematic review and meta-analysis focused on MRI surveillance



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BACKGROUND

Optic pathway gliomas (OPGs) are among the most frequent central nervous system tumors in children with neurofibromatosis type 1 (NF1), yet their prevalence has not been systematically synthesized. Pooled prevalence estimates are important to inform clinical surveillance practices, particularly given the critical role of early symptom detection in preserving vision and the debate regarding magnetic resonance imaging (MRI) surveillance in asymptomatic patients. The purpose of this study was to investigate the prevalence of OPG in children (<18 years) with NF1 based on pooled data in the literature and describe its variability.

METHODS

The databases of PubMed, Embase, Scopus, Web of Science, and Cochrane CENTRAL were searched without language or date restrictions. Eligible studies reported OPG prevalence in patients with NF1 <18 years of age. Two reviewers independently screened studies, extracted data, and assessed quality. Studies meeting minimum methodological quality were included in our meta-analysis, and subgroup and meta-regression analyses were performed.

RESULTS

A total of 38 studies encompassing 6,314 patients were included in the qualitative synthesis; 27 studies (5,485 patients) were subject to meta-analysis. The pooled prevalence of optic pathway gliomas was 17% (95% CI, 14%-20%), with low certainty according to GRADE. Subgroup analysis found no significant association in reported rates across MRI surveillance strategies, and a focused meta-regression directly comparing routine and symptom-based approaches showed similar results. Analyses by continent, sample size, and quality score revealed no significant differences.

CONCLUSIONS

Our results provide the first pooled evidence prevalence estimate of OPG in children and adolescents with NF1 and highlight the importance of close monitoring, reinforcing guidelines that favor a symptom-based MRI approach and suggesting no clear advantage of routine imaging. (J AAPOS 2025;29:104701)



Optic pathway gliomas (OPGs) are among the most common central nervous system tumors in children with neurofibromatosis type 1

(NF1), with prevalence estimates of 15%-20% frequently cited in narrative reviews.¹⁻⁴ These low-grade gliomas, predominantly pilocytic astrocytomas, can involve the intraorbital optic nerves, prechiasmatic optic nerves, chiasm, optic tracts, or optic radiations. When symptomatic, they may lead to progressive vision loss, ocular abnormalities, and, depending on tumor location, endocrine or neurological dysfunction.¹⁻⁴

NF1-associated OPGs have a variable clinical course, with some remaining asymptomatic, requiring no intervention, and others progressing and causing visual or systemic complications.¹⁻⁴ Multiple visual signs and symptoms have been identified as predictors of poor visual outcomes,⁵ and chemotherapy tends to be more effective at halting further visual deterioration than restoring lost function.¹⁻⁴ Together, these factors highlight the importance of early detection of symptomatic OPGs to support timely management and preserve vision.

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Data are provided in [Supplement 1](#) (available at [jaapos.org](#)). Analysis code is available from the corresponding author upon reasonable request.

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Current guidelines recommend annual ophthalmologic evaluations until age 8 years and at least every other year thereafter through age 18. Magnetic resonance imaging (MRI) is reserved for children with clinical signs suggestive of OPG.^{3,6,7} However, the role of routine MRI surveillance in asymptomatic children with NF1 remains controversial: some studies suggest it may improve clinical outcomes, whereas others have found no benefit and have raised additional concerns, including the need for repeated sedation and the added burden on families.¹

To date, no systematic review has specifically synthesized prevalence data for OPGs in children and adolescents with NF1, who represent the population most affected by these tumors and for whom informed surveillance strategies are particularly important. This study aimed to address this gap through a systematic review and meta-analysis, with particular focus on MRI surveillance strategies, given their clinical relevance for imaging practices and surveillance guidelines.

Materials and Methods

This systematic review was registered with the International Prospective Register of Systematic Reviews (PROSPERO; registration number CRD42024616270) and conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) guidelines.⁸⁻¹⁰ The study did not entail the collection of new data from human or animal subjects; thus, ethical approval was not required.

Searches were performed on the following databases: MEDLINE via PubMed and Scopus (November 20, 2024), Web of Science Core Collection (November 21, 2024); and Embase and Cochrane Central Register of Controlled Trials (CENTRAL) via Ovid (January 30, 2025). No language or date restrictions were applied. Further details of our search strategy are provided in [Table S1 \(Supplement 1, available at \[jaapos.org\]\(http://jaapos.org\)\)](#).

Duplicate records were partially removed using Rayyan's deduplication feature prior to screening.¹¹ No records were excluded by automation tools. Two independent reviewers screened titles, abstracts, and full-text articles, extracted data, and conducted quality appraisal. In cases of disagreement, a third investigator was consulted, and discrepancies were resolved by consensus.

To be included, studies had to be on humans and analyze the prevalence of OPG in children and adolescents (0-18 years of age) with NF1. Articles were excluded if they (1) were reviews and meta-analyses, case reports, editorials, letters, meeting abstracts, personal opinions or commentaries, or book chapters; (2) were experimental animal studies or in vitro studies; or (3) did not analyze the prevalence of OPG in children and adolescents aged 0-18 with NF1, or reported insufficient data for its calculation.

The following data were extracted into a standardized Excel spreadsheet: study information (ie, author, affiliation, publication year, location), number of patients with OPG, total number of patients with NF1, OPG prevalence (%), age at NF1 diagnosis, age at OPG diagnosis, methods for diagnosing NF1, methods for

diagnosing OPG, and MRI surveillance strategy. The methodological quality of included studies was assessed using the Joanna Briggs Institute (JBI) Quality Appraisal Checklist for Studies Reporting Prevalence Data.¹² This nine-item checklist (henceforth, Q1-Q9) evaluates the following parameters: (Q1) the sample frame was appropriate to address the target population; (Q2) participants were sampled appropriately; (Q3) the sample size was adequate; (Q4) study subjects and setting were described in detail; (Q5) data analysis was conducted with sufficient coverage of the identified sample; (Q6) valid methods were used to identify the condition; (Q7) the condition was measured in a standard and reliable way for all participants; (Q8) the statistical analysis was appropriate; (Q9) and the response rate was adequate, or if not, whether the low response rate was managed appropriately. Each item was rated as "yes," "no," "unclear," or "not applicable." Only studies scoring "yes" on at least five of the nine items were included in the meta-analysis, consistent with thresholds used in prior prevalence reviews,^{13,14} and to ensure inclusion of studies meeting a minimum standard of methodological quality.

A meta-analysis of proportions was performed using the meta-prop function from the meta package 8.2-1 (Balduzzi S, Rücker G, Schwarzer G [2019]) in R 4.4.2 (R Core Team [2021], R: A language and environment for statistical computing, R Foundation for Statistical Computing, Vienna, Austria, <https://www.R-project.org/>). Both common-effect and random-effects models were calculated, with primary focus placed on the random-effects estimates due to assumed heterogeneity among studies. The inverse variance method was used, and the Freeman-Tukey double arcsine transformation was applied to stabilize variances. Pooled prevalence estimates were reported with 95% confidence intervals, and a prediction interval was included to reflect the expected range of true effects in future studies.¹⁵ Sensitivity analyses were not conducted, as only studies meeting a minimum threshold for methodological quality were included in the meta-analysis. Heterogeneity was assessed using the I^2 statistic and Cochran's Q test. Forest plots were generated to visualize study-level and pooled estimates. Publication bias was evaluated qualitatively via funnel plot symmetry and quantitatively using Egger's test,¹⁶ implemented through the metabias function.

Subgroup analyses were conducted by MRI surveillance strategy and continent using the subgroup argument. The MRI surveillance strategy variable was categorized as "routine," "symptom-based," or "unclear/mixed," based on imaging approach used to detect OPGs in each study. Only studies that explicitly reported MRI as the surveillance modality were assigned to the routine or symptom-based categories. Studies that lacked sufficient information or used other diagnostic imaging approaches were classified as unclear/mixed. Subgroup-specific prevalence estimates were calculated, and between-group differences were tested using χ^2 statistics.

Two separate random-effects meta-regression approaches were used. Associations between OPG prevalence and study sample size and quality appraisal score were examined using the metareg function (meta package) in R 4.4.2, applied to the pooled prevalence model. Results were visualized using scatter plots with fitted regression lines. A separate meta-regression comparing routine versus symptom-based MRI surveillance strategies was conducted

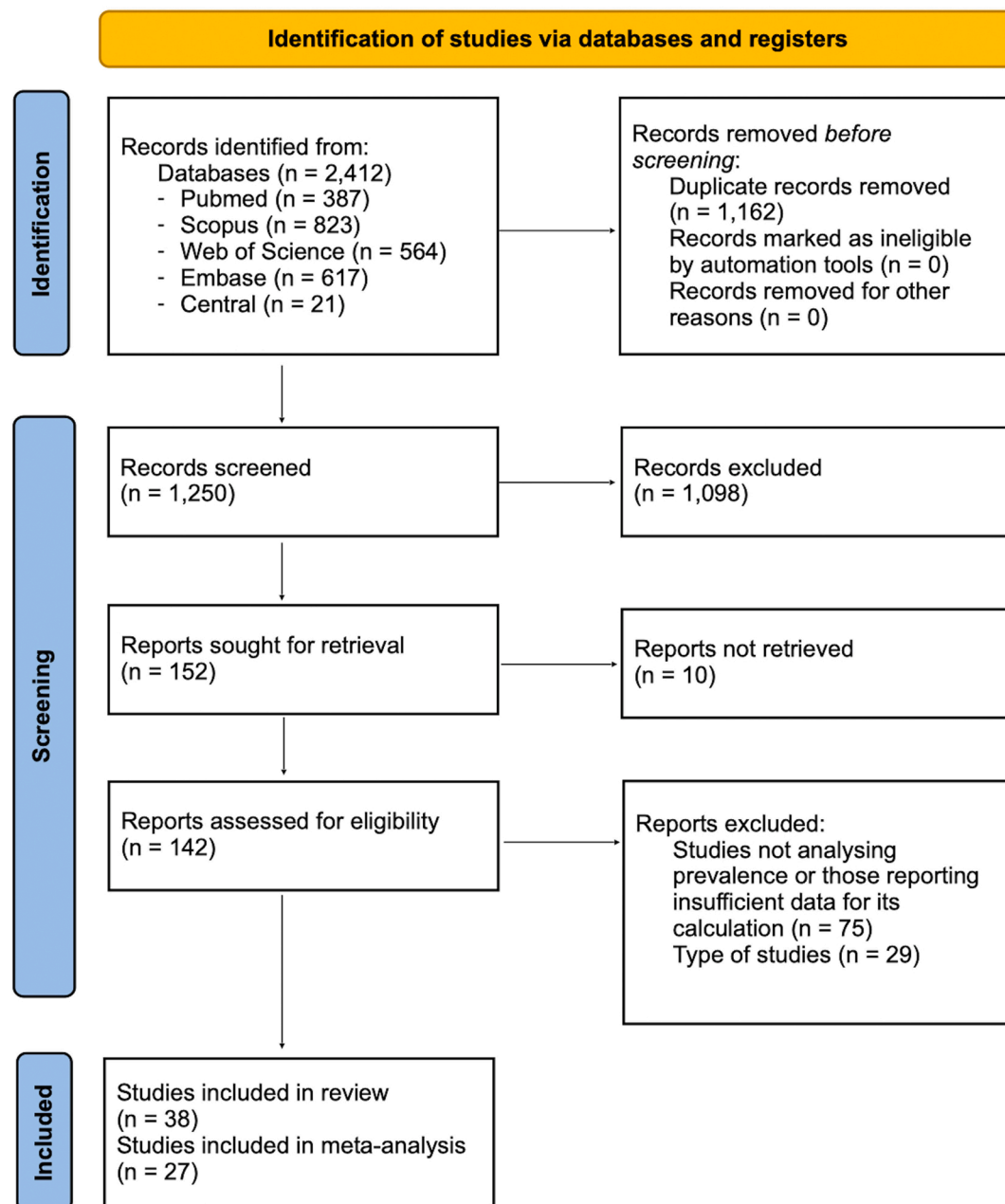


FIG 1. PRISMA 2020 flow diagram of study selection. The flow diagram illustrates the number of records identified, screened, excluded, and included in the systematic review and meta-analysis.

using the *rma* function from the *metafor* package 4.8-0 (Viechtbauer W [2010]), with model estimates reported on the logit scale and visualized using dot-and-whisker plots.

We assessed the certainty of evidence for the primary outcome using the GRADE (grading of recommendations, assessment, development and evaluation) approach,¹⁷ as outlined by the Cochrane Effective Practice and Organisation of Care group.¹⁸ Certainty of evidence was judged across five domains: risk of bias, inconsistency, indirectness, imprecision, and publication bias. Each domain was rated as not serious, serious, or very serious.

Observational studies of prevalence start at a high certainty level and may be downgraded based on concerns in any of the domains.

Results

The electronic database search and screening process yielded a total of 38 studies that met the inclusion criteria. All 38 studies were included in the qualitative synthesis, of which 27 met the quality threshold for inclusion in the quantitative meta-analysis. The PRISMA flow diagram

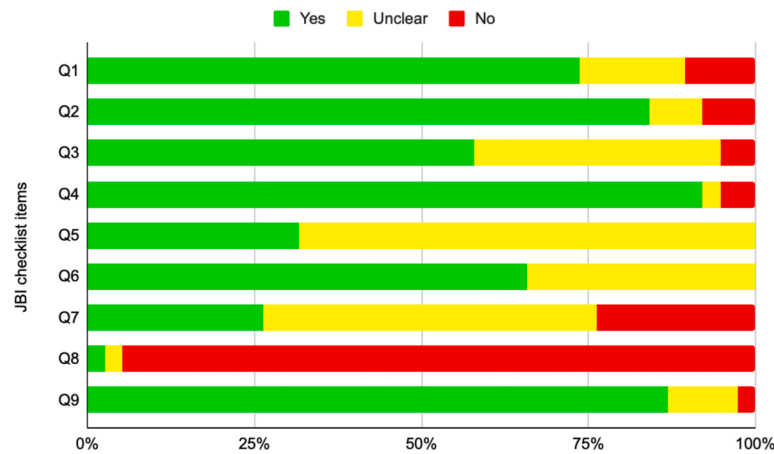


FIG 2. Summary of methodological quality for included studies. Proportion of studies rated “yes,” “unclear,” or “no” for each of the nine Joanna Briggs Institute (JBI) checklist items for studies reporting prevalence data. Item definitions are provided in the Methods section and [Table S3 \(Supplement 1\)](#).

summarizing the study selection process is presented in [Figure 1](#).

Study Characteristics

The 38 included studies were published between the mid-1980s and 2024, reflecting approximately four decades of research on NF1-associated OPGs. In total, these studies encompassed 6,314 children and adolescents with NF1, with individual study sample sizes ranging from 27 to 826 NF1 patients. The studies were conducted across multiple continents, including Europe, North America, Asia, Oceania, and Africa, indicating broad geographic representation.

The majority were retrospective cohort or cross-sectional designs based on hospital or registry datasets. Most studies employed either National Institutes of Health (NIH) criteria, genetic testing, or both for NF1 diagnosis, while OPG diagnosis was most commonly based on MRI, although some studies lacked explicit reporting. MRI surveillance strategies varied across studies, with 11 using routine MRI, 6 using symptom-based MRI, and the remainder either lacking sufficient information or using other diagnostic modalities.

The mean or median age at NF1 diagnosis ranged from approximately 2 to 7.7 years, and age at OPG diagnosis from 2.2 to 7.4 years, although age data were inconsistently reported across studies. Prevalence estimates for OPG in NF1 patients ranged from 4% to 56%, indicating substantial variability across the included studies. Detailed study-level characteristics, including reference numbers, are summarized in [Table S2 \(Supplement 1\)](#). Full bibliographic details for included studies are provided in the references of [Supplement 1](#).

Methodological Quality

Methodological quality was assessed using the Joanna Briggs Institute (JBI) checklist. The nine checklist items

(Q1–Q9) are defined in the Methods section and detailed in [Table S3 \(Supplement 1\)](#). [Figure 2](#) summarizes the proportion of studies rated “yes,” “unclear,” or “no” for each item.

Most studies showed low risk of bias across several domains, with 75% to 100% of them meeting criteria for appropriate participant sampling, clearly described study settings, and adequate response rates (Q2, Q4, Q9). Between 50% and 75% of studies were rated “yes” for having an appropriate sampling frame, adequate sample sizes, and valid diagnostic methods for OPG (Q1, Q3, Q6).

Methodological limitations were most evident in three specific areas. Only about 30% of studies clearly reported sufficient coverage of the identified NF1 population (Q5) or used consistent and standardized methods to measure optic pathway glioma (Q7). Both domains also had the highest proportion of “unclear” ratings, reflecting limited reporting. The most significant concern was statistical analysis (Q8), which was adequately addressed in only one study.

A detailed, study-level quality assessment is presented in [Figure S1 \(Supplement 1\)](#), including overall quality scores and meta-analysis inclusion status.

Pooled Prevalence of OPG in NF1

The 27 studies in the meta-analysis contributed data on 5,485 NF1 patients and 968 OPG cases. The pooled prevalence of OPG in children and adolescents under 18 years of age with NF1 was 17% (95% CI, 14%–20%) based on a random-effects model ([Figure 3](#)).

A high degree of heterogeneity was observed across the included studies, as indicated by an I^2 value of 80.3%, which exceeds the threshold typically considered substantial according to Higgins’s classification.¹⁹ The 95% prediction interval ranged from 7% to 31%, reflecting the

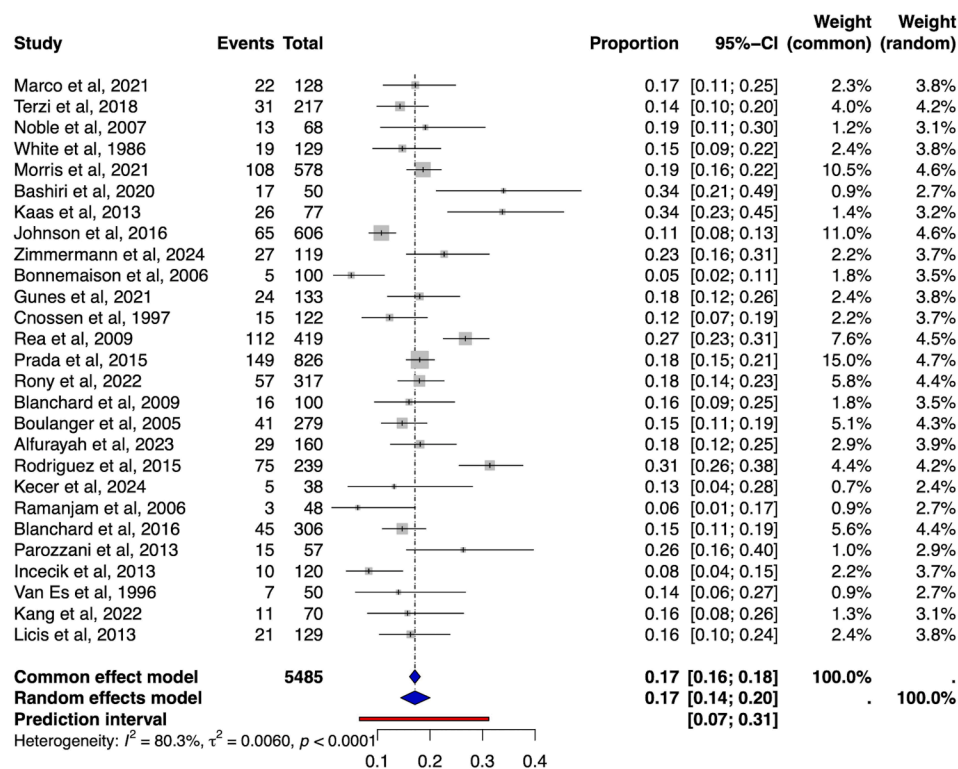


FIG 3. Forest plot of pooled optic pathway gliomas (OPG) prevalence in children and adolescents with NF1. Individual and pooled prevalence estimates with 95% confidence intervals based on 27 studies.

expected variation in prevalence rates among comparable future studies. Sensitivity analyses were not conducted due to the inclusion of studies with minimum methodological quality.

MRI Surveillance Strategy and OPG Prevalence

Subgroup analysis based on MRI surveillance strategy (Figure 4) showed that the pooled prevalence of OPG was 15% (95% CI, 11%-21%) in studies using Routine surveillance, 19% (95% CI, 12%-27%) in studies using Symptom-based surveillance, and 18% (95% CI, 13%-23%) in studies categorized as unclear/mixed, which included non-MRI or unreported imaging approaches. Despite these numerical differences, the variation between subgroups was not statistically significant ($P = 0.567$). Heterogeneity within each subgroup remained substantial ($I^2 = 78\%$ – 83%), suggesting that surveillance strategy alone did not account for the between-study variability.

To directly compare the two clearly defined MRI surveillance strategies, a meta-regression was performed, including only studies classified as using routine or symptom-based surveillance (Supplement 1, Figure S2). This analysis found no statistically significant association between the two MRI surveillance strategies and OPG prevalence ($\beta = -0.22$; 95% CI, -0.73 to 0.30 ; $P = 0.413$), reinforcing the finding that there was no apparent detection advantage associated with routine MRI.

Further Exploration of Heterogeneity

To explore potential drivers of heterogeneity beyond MRI surveillance strategy, we conducted meta-regression analyses and additional subgrouping by continent. Meta-regression revealed no statistically significant association between prevalence and study sample size ($P = 0.993$; see Supplement 1, Figure S3). Similarly, no significant association was found with quality appraisal scores, although a trend toward lower prevalence in higher-quality studies was observed ($P = 0.094$; see Supplement 1, Figure S4). Subgroup analysis by continent (Europe, North America, Asia, Oceania, and Africa) also did not identify statistically significant differences in prevalence ($P = 0.211$; see Supplement 1, Figure S5).

Publication Bias

Visual inspection of the funnel plot (Supplement 1, Figure S6) revealed a fairly symmetrical distribution, suggesting no strong evidence of small-study effects. This was supported by Egger's regression test, which showed no evidence of publication bias ($P = 0.974$).

Certainty of the Evidence

The certainty of the evidence for the pooled prevalence estimate was assessed as very low according to the GRADE approach (Supplement 1, Table S4). This rating was downgraded across three domains. Inconsistency was present

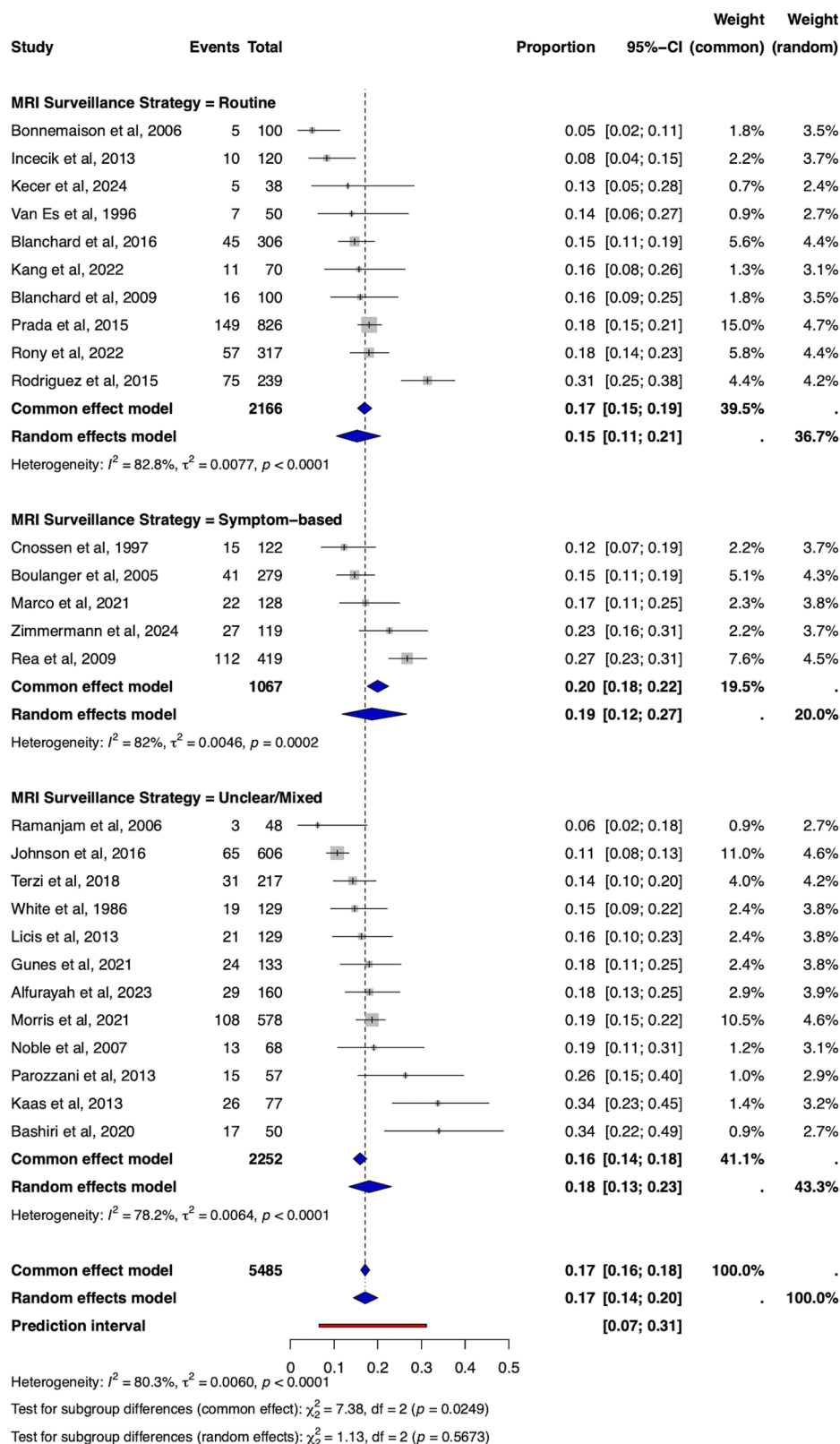


FIG 4. OPG prevalence stratified by magnetic resonance imaging (MRI) surveillance strategy. Forest plot of subgroup estimates for routine, symptom-based, and unclear/mixed MRI surveillance strategies. The unclear/mixed category includes studies with insufficient imaging information or those using non-MRI modalities. No statistically significant difference was found between subgroups ($\chi^2 = 1.13$; $df = 2$; $P = 0.567$; random-effects model).

due to substantial heterogeneity ($I^2 = 80.3\%$) that could not be explained by subgroup or meta-regression analysis. Indirectness was noted because several included studies did not explicitly aim to estimate prevalence, instead reporting it as a secondary outcome. Imprecision was reflected in the wide prediction interval (7%-31%), despite a relatively narrow confidence interval (14%-20%). There were no serious concerns regarding risk of bias or publication bias. All meta-analysed studies scored ≥ 5 on the JBI checklist, and both funnel plot symmetry and Egger's test ($P = 0.974$) supported the absence of small-study effects.

Discussion

To our knowledge, this is the first systematic review and meta-analysis to synthesize prevalence estimates of OPGs in children and adolescents <18 years of age with NF1. We identified a pooled prevalence of 17% (95% CI, 14%-20%), falling within the commonly cited range of 15%-20%.¹⁻⁴ This finding, together with the combination of clinical variability,¹⁻⁴ the negative prognostic value of multiple visual symptoms,⁵ and the fact that chemotherapy is generally more effective at halting further deterioration than restoring vision,¹⁻⁴ supports the rationale for close monitoring of visual function as part of NF1 follow-up.

Subgroup analysis showed a lack of association between MRI surveillance strategy and OPG prevalence. The pooled prevalence was 15% (95% CI, 11%-21%) among studies using routine surveillance and 19% (95% CI, 12%-27%) among those using symptom-based surveillance, showing close numerical estimates and overlapping confidence intervals. To further explore this relationship, a meta-regression restricted to studies with clearly defined MRI surveillance strategies was performed, excluding studies with mixed or unclear approaches. This analysis, comparing routine and symptom-based MRI strategies, similarly found no statistically significant difference in OPG prevalence ($\beta = -0.22$; 95% CI, -0.73 to 0.30 ; $P = 0.413$). Together, these results provide new evidence against a detection advantage associated with routine MRI surveillance and reinforce current clinical guidelines discouraging routine MRI screening in asymptomatic children with NF1.⁶

Other potential sources of heterogeneity, including sample size, geographic region, and methodological quality, were explored through subgroup and meta-regression analyses. No statistically significant associations were observed. Of note, studies of relatively higher methodological quality tended to report lower prevalence, which may reflect the use of stricter diagnostic or inclusion criteria.

Key strengths of this review include a preregistered protocol, comprehensive and unrestricted literature search, independent screening and appraisal by multiple reviewers, and adherence to PRISMA and MOOSE guidelines. We also explored heterogeneity and assessed certainty of evidence using the GRADE approach.

Several limitations must be acknowledged. Considerable heterogeneity remained unexplained across studies, possibly reflecting differences in study design, diagnostic definitions, and timing for imaging. Moreover, the certainty of evidence was rated very low according to the GRADE approach, due to inconsistency, indirectness, and imprecision. In addition, prevalence estimates alone do not address when treatment should be considered or how surveillance affects visual outcomes, with both being essential to shaping surveillance decisions. Lastly, excluding lower-quality studies from the meta-analysis may improve validity but could introduce selection bias.

Overall, this study provides the first pooled estimate of OPG prevalence in children and adolescents with NF1 and reinforces the value of a symptom-based MRI surveillance approach. Along with these contributions, the presence of unexplained heterogeneity across studies, the very low certainty of evidence according to GRADE, and the absence of outcome data analysis should be noted. Future research should follow prospective NF1 cohorts using consistent methods, more detailed age-at-diagnosis reporting, and clinically relevant outcomes to better quantify OPG prevalence, explain heterogeneity, and guide imaging decisions.

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